



REPORT

System Hardening via Local GPO: Restricting Anonymous Connections

v1.0.0

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REVISION HISTORY

Version	Date	 Author	Description of Changes
v1.0.0	09/30/2025	Eldon G.	Initial draft.





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I. INTRODUCTION AND PROJECT SCOPE

1.1 Project Description

This project was performed on a Windows 10 computer using the Local Group Policy (`gpedit.msc`). The aim is to stop unknown users from accessing system details without permission.



Disclaimer: This report documents my personal work completing an MCSI lab exercise. It reflects my understanding and configuration of Windows 10 Local Group Policy settings for operating system patching in a controlled, offline environment. No MCSI instructional videos, lab guides, or proprietary materials have been posted, shared, or distributed. The content has been written independently to demonstrate my skills while remaining fully compliant with MCSI's academic pledge and policies.



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1.2 Security Goals

The goal is to make the system safer by stopping features that allow unknown users to use system resources without logging in.

1.3 Objectives

1. Disable anonymous SID/Name translation.
2. Block enumeration of SAM accounts and shares.
3. Restrict anonymous access to named pipes and shared resources.
4. Deny network access to local accounts.
5. Enforce secure client/server authentication.

1.4 Portfolio Value

This work shows skill in making Windows more secure, managing the Group Policy, and protecting systems from being spied on.



II. THREAT BACKGROUND AND RATIONALE

2.1 Threat of Anonymous Connections (Null Sessions)

A null session is a legacy feature that allows remote connections without credentials. Attackers exploit null sessions to enumerate system information without any authentication.

2.2. Adversary Reconnaissance (the “Why”)

Through anonymous connections, attackers can:

- Enumerate the users and groups.
- Translate SIDs into usernames.
- Shared resources and named pipes were discovered.
- Gather OS and policy information on targeted attacks.

2.3. Control Mechanism

Disabling anonymous access via the Local Group Policy enhances security. This aligns with the CIS Benchmarks and Microsoft standards.



III. GPO IMPLEMENTATION

3.1. Implementation Environment

The configuration was performed on a Windows 10 workstation using the **Local Group Policy Editor** (`gpedit.msc`).

3.2. Configuration Summary: Anonymous Access Restrictions

A series of security policies was applied to remove the ability for unauthenticated users to perform reconnaissance against the system. These controls focused on:

1. **Disabling translation and enumeration features** that allow discovery of local accounts, groups, and system identifiers.
2. **Restricting insecure network behaviors**, including legacy fallback mechanisms and permissive access models.
3. **Enforce access denial rules** for non-privileged accounts, ensuring local accounts cannot be abused for remote sessions.

3.3. Policy Grouping

In total, **eleven policies** were set across the *Security Options*, *Administrative Templates*, and *User Rights Assignment* categories. Collectively, these policies blocked anonymous SID/name translation, prevented enumeration of SAM accounts and shares, restricted insecure NTLM usage, and denied unauthorized accounts from connecting remotely.

This layered approach eliminated the primary pathways for null session exploitation without impacting normal authenticated system use.



IV. Technical Verification and Proof of Work

4.1. Policy Application

After configuration, the settings were enforced immediately with the `gpupdate /force` command to ensure all changes were active.

4.2. Verification Method

Validation was performed using a combination of:

- **Registry queries** to confirm that underlying system values had been updated to match the intended security posture.
- **Enumeration attempts** (e.g., user and share listing tests) to demonstrate that anonymous connections no longer produce results.

4.3. Demonstration of Hardening

Before configuration, anonymous sessions could return account or share data; these queries were now blocked, showing that the system successfully enforced the new restrictions. Registry checks confirmed that the relevant keys reflected the hardened state.

4.4. Conclusion of Verification

The verification process confirmed that the workstation was operating under a **hardened baseline**, resistant to reconnaissance through unauthorized anonymous connections.



V. CONCLUSION

5.2 Key Takeaways

All necessary Local Group Policy settings were set up. This prevented the system details from being listed anonymously.

Learning Outcomes

- Direct link between Group Policy settings and underlying registry configuration.
- Understanding how system baselines mitigate adversary reconnaissance techniques.

5.2 Security Implications and Recommendations

- Monitor system security logs for failed anonymous logon attempts.
 - Hardening is strengthened by adding identity rules, limiting NT LAN Manager (NTLM), and using Kerberos authentication.
 - Replicate the configuration across enterprise systems using domain-level GPOs.
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